valid along every solution of Eqs. (10). These inequalities insure piecewise monotonic time variations of V,γ, and h, whereas energy considerations establish the boundedness of all three quantities when p_0 satisfies Eq. (9), thus insuring the existence of the limit $P_{p_0}{}^*$.

In the second phase of the proof, it is shown that $P_{p_0}{}^* = (0,\pi,0)$, whenever this limit exists. The identity,

$$\frac{d}{dt}\ln\left(\frac{V^2}{h}\right) = -\frac{2\mu}{(R+h)^2V}\left[1+\frac{V^2(R+h)^2}{4\mu h}\right]\cdot \left[\cos\gamma + \left\{\cos^2\gamma + \frac{\sin^2\gamma}{1+[V^2(R+h)^2/4\mu h]}\right\}^{1/2}\right] \quad (13)$$

is instrumental in this connection and leads to the following bounds on the variation of V^2/h:

$$V_0{}^2/h_0 \geq V^2/h \geq \epsilon_0(V_0{}^2/h_0) \quad (14)$$

with $1 \geq \epsilon_0 > 0$ and ϵ_0 dependent on p_0 in general. These inequalities, together with a few standard results from the theory of ordinary differential equations, lead directly to the conclusion $P_{p_0}{}^* = (0,\pi,0)$. The arguments in this stage are facilitated by treating γ in place of t as an independent variable on terminal subarcs; this is permissible in view of the monotonic behavior of γ established earlier.

The final phase of the proof demonstrates the finiteness of $T_{p_0}{}^*$ whenever the limit $P_{p_0}{}^*$ exists; here it is convenient (and permissible) to employ h as an independent variable on terminal subarcs.

To summarize, it has been shown that a constant thrust deceleration formula based upon a flat planet, uniform g-field approximation is generally capable of producing perfect soft landings with spherical planet, inverse square g-field dynamics, provided the formula is implemented in a continuous closed-loop fashion (there are certain peculiar "hyperbolic" initial conditions for which the statement is not true, but these have little practical significance). This means that the closed-loop implementation of the local-g version of formula (5) is self-correcting in the sense that it causes the vehicle's state vector to move into regions where the errors due to the underlying approximations in Eq. (5) diminish with sufficient rapidity to guarantee the ultimate satisfaction of the soft landing boundary condition (3). It does not mean that the closed-loop implementation nullifies other persistent sources of error, e.g., state vector estimation errors and control variable actuation errors. These important questions lie beyond the present investigation.

There is also the question of fuel consumption penalties incurred in the feedback implementation. The second author has compared fuel consumption figures for the closed-loop controller (obtained by numerical integration) against those for constant thrust deceleration descents (found through iterative trial and error procedures) and also against those for idealized one and/or two tangent impulse maneuvers. Over the range of initial conditions considered, the feedback controller is equal or superior to the constant thrust deceleration descent and nearly as good as the impulsive maneuver (within 1–2% for the "circular" initial states tested). In particular, the feedback controller is never accompanied by a tendency toward total mass consumption near any point in the finite region of the state space; this contrasts with constant thrust deceleration descents that always display this tendency near critical points from which such descents are impossible. A general proof of this statement credited to the second author is available in Ref. 3, but cannot be presented here because of space limitations. A number of curves summarizing the results of the numerical simulation work also appear in Ref. 3.

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Rates of Change of Eigenvalues and Eigenvectors

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Introduction

THE design of complex structures to satisfy dynamic response restrictions is hampered by the inherent difficulty and the computational cost of dynamic analysis. These limitations put a low ceiling on the number of trial designs which can be analyzed. Furthermore, from the analysis alone, it is not usually clear how a design should be modified to improve or maintain its dynamic properties. These two problems are accentuated when the structural design process involved is an automated optimum design procedure. In these computer design methods (for example, the method of feasible directions), there is a fundamental requirement for information relating changes in the dynamic response quantities to changes in the design variables as well as a need for rapid reanalysis of modified trial designs.

A significant part of the dynamic analysis involves the eigensolution of the problem

$$[K]\mathbf{X} = \lambda[M]\mathbf{X} \quad (1)$$

where the scalar quantities λ (eigenvalues) and the corresponding nontrivial vectors $\mathbf{X}$ (eigenvectors) are to be determined. In the common structural application, the $n \times n$ matrices $[K]$ and $[M]$ are, respectively, the master stiffness and mass matrices of the structure; their order n corresponds to the number of elastic degrees of freedom of the system. These are symmetric matrices, the components of which are dependent on the design parameters.

This Note presents exact expressions for the rates of change of eigenvalues and eigenvectors with respect to the design parameters of the actual structure, and indicates that these derivatives can be used successfully to approximate the analysis of new designs.

Suppose λ_i is the eigenvalue corresponding to ith natural frequency of a design $\boldsymbol{\delta}$ where $\boldsymbol{\delta}$ denotes the r-component vector of design variables for the structure. Suppose further that the design variables undergo a change of $\Delta\boldsymbol{\delta} \equiv (\Delta\delta_1,\Delta\delta_2, \ldots \Delta\delta_r)$. An approximation to the ith eigenvalue of the new design $\boldsymbol{\delta}^*$ is given by

$$\lambda_i{}^* \simeq \lambda_i + \Delta\boldsymbol{\delta}^T\nabla\lambda_i \equiv \lambda_i{}^{(E)} \quad (2)$$

where $\nabla\lambda_i$ represents the vector $(\lambda_{i,1},\lambda_{i,2},\ldots, \lambda_{i,r})$ and $\lambda_{i,j}$ denotes $\partial\lambda_i/\partial\delta_j$. Furthermore, if $\mathbf{X}_i$ is the associated

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eigenvector, then

$$\mathbf{X}_i^* \simeq \mathbf{X}_i + [\nabla \mathbf{X}_i]\Delta\delta \equiv \mathbf{X}_i^{(E)} \tag{3}$$

where the $n \times r$ matrix

$$[\nabla \mathbf{X}_i] \equiv [\mathbf{X}_{i,1}, \mathbf{X}_{i,2}, \ldots, \mathbf{X}_{i,r}]$$

and $\mathbf{X}_{i,j}$ denotes $\partial \mathbf{X}_i/\partial \delta_j$. Using the Rayleigh quotient' another approximation to λ_i^* is given by

$$\lambda_i^* \simeq \frac{\mathbf{X}_i^{(E)T}[K^*]\mathbf{X}_i^{(E)}}{\mathbf{X}_i^{(E)T}[M^*]\mathbf{X}_i^{(E)}} \equiv \lambda_i^{(R)} \tag{4}$$

where $[K^*] \equiv [K(\boldsymbol{\delta}^*)]$ and $[M^*] \equiv [M(\boldsymbol{\delta}^*)]$.

Rate of Change of an Eigenvalue

Let

$$[F_i] = [F(\lambda_i, \boldsymbol{\delta})] \equiv [K - \lambda_i M] \tag{5}$$

where λ_i is ith eigenvalue; therefore, from Eq. (1), we obtain

$$[F_i]\mathbf{X}_i = 0 \tag{6}$$

Premultiplication of Eq. (6) by $\mathbf{X}_i^T$ gives

$$\mathbf{X}_i^T[F_i]\mathbf{X}_i = 0 \tag{7}$$

Differentiating Eq. (7) with respect to δ_j, we get

$$\mathbf{X}_{i,j}[F_i]\mathbf{X}_i + \mathbf{X}_i^T[(\partial F_i/\partial \delta_j)]\mathbf{X}_i + \mathbf{X}_i^T[F_i]\mathbf{X}_{i,j} = 0 \tag{8}$$

The first and the third term of Eq. (8) are zero, by virtue of Eq. (6), and because $[F_i]$ is a symmetric matrix. Therefore, we obtain

$$\mathbf{X}_i^T \left[\frac{\partial F_i}{\partial \delta_j}\right]\mathbf{X}_i = 0 \tag{9}$$

Differentiating Eq. (5) we obtain

$$(\partial F_i/\partial \delta_j) = [K_{,j} - \lambda_i M_{,j} - \lambda_{i,j} M] \tag{10}$$

where $[K_{,j}]$ and $[M_{,j}]$ denote the matrices formed by differentiating the elements of $[K]$ and $[M]$ matrices, respectively, with respect to δ_j. These matrices are presumably readily computed and they are generally quite sparse.

If the eigenvectors are taken to be M-orthonormal, i.e.,

$$\mathbf{X}_i^T[M]\mathbf{X}_i = 1 \tag{11}$$

then from Eqs. (9–11) we obtain

$$\lambda_{i,j} = \mathbf{X}_i^T[K_{,j} - \lambda_i M_{,j}]\mathbf{X}_i \tag{12}‡$$

Note that the expression of Eq. (12) involves only the eigenvalue and eigenvector under consideration, and thus a complete solution of eigenproblem is not needed to obtain these derivatives.

Rate of Change of an Eigenvector

Formulation 1

Differentiating the vector Eq. (6), we obtain

$$[F_i]\mathbf{X}_{i,j} = -[F_{i,j}]\mathbf{X}_i \tag{13}$$

where $F_{i,j} \equiv \partial F_i/\partial \delta_j$ is given by Eq. (10). Also differentiating Eq. (11) we obtain the scalar equation

$$2\mathbf{X}_i^T[M]\mathbf{X}_{i,j} = -\mathbf{X}_i^T[M_{,j}]\mathbf{X}_i \tag{14}$$

Note that $[F_i]$ is a singular matrix of rank $(n - 1)$ if the system of equations represented by Eq. (1) has distinct eigenvalues. Further, it can be shown that Eq. (14) is linearly independent of the equations represented by Eq. (13). Taking these together, we obtain a consistent but overdetermined system, which can be written in partitioned form as

$$\underset{[(n+1)\times n]}{\left[\begin{array}{c} F_i \\ \hdashline 2\mathbf{X}_i^T[M] \end{array}\right]} \underset{(n\times 1)}{\mathbf{X}_{i,j}} = -\underset{[(n+1)\times n]}{\left[\begin{array}{c} F_{i,j} \\ \hdashline \mathbf{X}_i^T[M_{,j}] \end{array}\right]} \underset{(n\times 1)}{\mathbf{X}_i} \tag{15}$$

Premultiplying Eq. (15) by $[(F_i/\mathbf{X}_i^T[M])]^T$ we can obtain the desired expression

$$\mathbf{X}_{i,j} = -[F_iF_i + 2M\mathbf{X}_i\mathbf{X}_i^TM]^{-1}[F_iF_{i,j} + M\mathbf{X}_i\mathbf{X}_i^TM_{,j}]\mathbf{X}_i \tag{16}$$

Once again the expression of Eq. (16) to find the rate of change of an eigenvector with respect to any design parameter involves only the eigenvalue and eigenvector under consideration and other known quantities.

Formulation 2

Since the eigenvectors form a complete set of vectors, any n-component vector can be represented as a linear combination of these, and, in particular, $\mathbf{X}_{i,j}$ can be represented as

$$\mathbf{X}_{i,j} = \sum_{k=1}^{n} a_{ijk}\mathbf{X}_k \tag{17}$$

Substituting Eq. (17) into Eq. (13) yields

$$[F_i]\sum_{k=1}^{n} a_{ijk}\mathbf{X}_k = -[F_{i,j}]\mathbf{X}_i \tag{18}$$

‡ A similar expression is available in Refs. 1 and 2. Furthermore, in Ref. 3, an identical expression has been used for the elementary error analysis of the eigenproblem arising in structural dynamics.

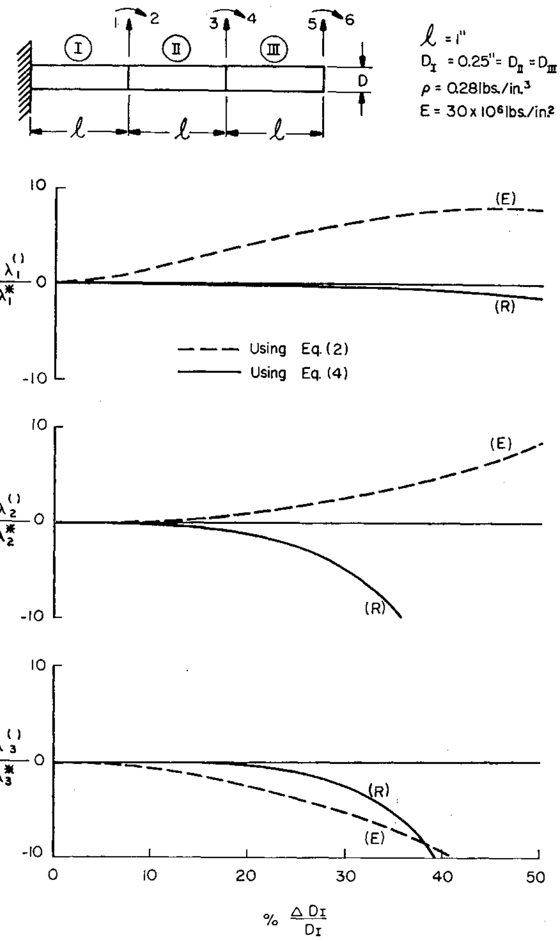

Fig. 1 Percentage error in prediction of λ_1, λ_2, and λ_3 of the perturbed design (root diameter increased).

Table 1 **Eigenvalues and rate of change of eigenvalues with respect to D_{I}, D_{II}, and D_{III}**

Number (i)	Eigenvalue λ_i	$\lambda_{i,\mathrm{I}}$	$\lambda_{i,\mathrm{II}}$	$\lambda_{i,\mathrm{III}}$
1	0.2466×10^8	0.3209×10^9	0.3456×10^8	-0.1582×10^9
2	0.9747×10^9	0.3860×10^{10}	0.4352×10^{10}	-0.4144×10^9
3	0.7782×10^{10}	0.2350×10^{11}	0.1709×10^{11}	0.2167×10^{11}

Table 2 **Rate of change of eigenvectors with respect to D_{I}: Components of $X_{i,\mathrm{I}} \times 10^{-2}$**

Eigenvalue no. (i)	$x_{1i,\mathrm{I}}$	$x_{2i,\mathrm{I}}$	$x_{3i,\mathrm{I}}$	$x_{4i,\mathrm{I}}$	$x_{5i,\mathrm{I}}$	$x_{6i,\mathrm{I}}$
1	−0.962	1.745	−0.668	−1.708	1.478	−2.298
2	2.644	5.737	−1.801	0.974	0.057	−3.046
3	−3.005	−25.78	−0.195	7.359	0.335	−5.307

and premultiplying both sides of Eq. (18) by $\mathbf{X}_l^T$, $l \neq i$ gives

$$\sum_{k=1}^{n} a_{ijk}\mathbf{X}_l^T[F_i]\mathbf{X}_k = -\mathbf{X}_l^T[F_{i,j}]\mathbf{X}_i \qquad (19)$$

wherefrom we obtain

$$a_{ijl} = \frac{\mathbf{X}_l^T[K_{,j} - \lambda_i M_{,j}]\mathbf{X}_i}{(\lambda_i - \lambda_l)} \qquad l \neq i \qquad (20)$$

since the M-orthonormal vectors are also K-orthogonal. In order to get a_{iji}, we substitute Eq. (17) into Eq. (14), which gives

$$2\mathbf{X}_i^T[M]\sum_{k=1}^{n} a_{ijk}\mathbf{X}_k = -\mathbf{X}_i^T[M_{,j}]\mathbf{X}_i \qquad (21)$$

Using the M-orthonormal relationship of the eigenvectors we get

$$a_{iji} = -(\mathbf{X}_i^T[M_{,j}]\mathbf{X}_i/2) \qquad (22)$$

It is to be noted that, in contrast with Eq. (16), the expression of Eq. (17), where the coefficients are given by Eqs. (20) and (22), involves the complete solution of an eigenproblem, but does not require the inversion of an additional $n \times n$ matrix. On the other hand, it is natural to speculate that, as is often done with the solution to the dynamics problem itself, it should be possible to approximate the derivative of the eigenvector in Eq. (17) by a partial sum

$$\mathbf{X}_{i,j} \simeq \sum_{k=1}^{r} a_{ijk}\mathbf{X}_k \qquad r < n$$

This possibility is being explored at this writing, but current results are inconclusive.

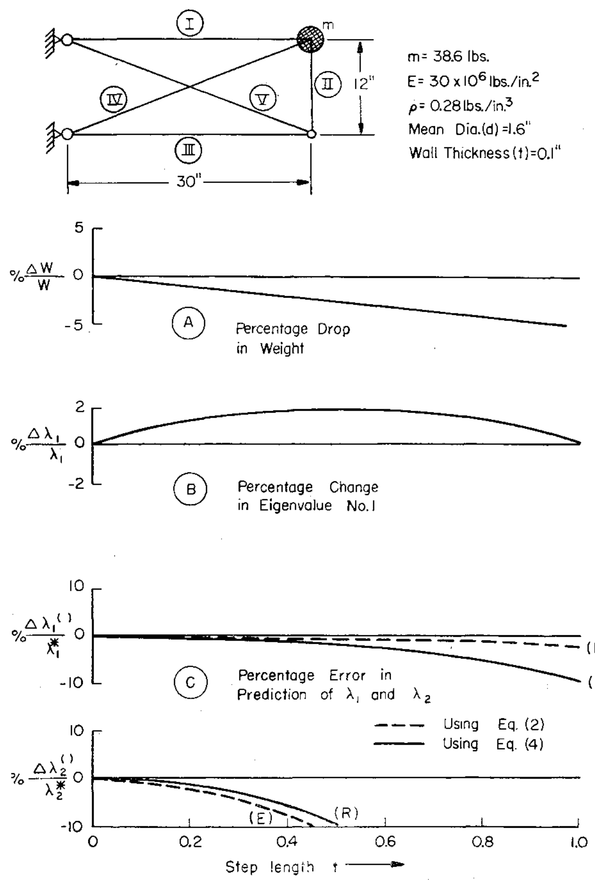

Fig. 2 Results for pin-connected planar frame.

Numerical Examples

Example a

As a simple application, the cylindrical cantilever beam shown in Fig. 1 is considered. It is modeled with three discrete elements; hence the dynamic analysis is approximated by six degrees of freedom. The distributed mass of the system has been used to evaluate the components of the mass matrix ("consistent" mass matrix). The design variables are the diameters of each element. In order to provide a check for other workers, results obtained for the rate of change of eigenvalues and eigenvectors are given in Tables 1 and 2.

The percentage error in approximating the first three eigenvalues of the perturbed design by increasing the root diameter D_I are shown in Fig. 1. The dashed curves represent the extrapolated values obtained from Eq. (2), whereas the continuous curves are those from the Rayleigh quotient expression of Eq. (4). Similar results were obtained when the middle or the free end diameters were changed.

For the lowest three frequencies of the beam, the extrapolated eigenvalues and extrapolated Rayleigh quotients are all reasonably accurate up to about 30% change in the root diameter. The Rayleigh quotients are usually the more accurate, but not universally. A possible indication as to when the approximations are breaking down is the divergence of $\lambda^{(E)}$ from $\lambda^{(R)}$.

Example b

Consider as another illustrative example the pin-connected planar "frame" shown in Fig. 2 consisting of tubular members. Each member is modeled as a beam element having 6 degrees of freedom; thus the system has 14 degrees of freedom. The mean diameter of each member has been taken as a design variable. A change vector of these five design variables was taken to be $\Delta\boldsymbol{\delta} \equiv t\{0.5, -1.0, -0.5, 0.2,$

$-0.2\}$ where t is a scalar representing the "length" of the step.

Results obtained by introducing this change in the design variables in steps of $t = 0.1, 0.2, \ldots, 1.0$ are plotted in Fig. 2. A 5% lowering of the weight was brought about as shown in Fig. 2, graph A, without lowering the lowest natural frequency of the structure (graph B). A plot of percentage error in the prediction of the first two eigenvalues is shown in graphs C of Fig. 2.

The sequential design change lowers the lowest individual natural frequencies of the elements II, III, and V at each step. The interaction between the falling member frequencies and the over-all structure frequency eventually causes a drop in the latter. The drastic change in the vibration mode which accompanies this phenomenon partly explains the loss of accuracy of $\lambda_1^{(R)}$ relative to $\lambda_1^{(E)}$ as seen in Fig. 2, graph C.

Conclusions

The exact expressions obtained for the rate of change of eigenvalues and eigenvectors, apart from their own usefulness in estimating the analysis of modified designs, are expected to be helpful in the automated optimum design of structures under dynamic response restrictions.

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Cylindrical Afterbodies at $M_\infty = 2$ with Hot Gas Ejection

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Nomenclature

A	= cross-sectional area of afterbody
d	= afterbody diameter
I	= ejection parameter $\dot{m}/\rho_\infty U_\infty A$
M_∞	= freestream Mach number
$\dot{m}$	= rate of ejection
p_b	= base pressure
p_∞	= freestream pressure
$\overline{(p_b/p_\infty)}$	= dimensionless base pressure increase $\dfrac{(p_b/p_\infty)_T - (p_b/p_\infty)_{\bar{T}=0}}{(p_b/p_\infty)_{\bar{T}=0}}$
T_g	= ejected gas temperature
T_t	= freestream stagnation temperature
$\bar{T}$	= $(T_g - T_t)/T_t$
U_∞	= freestream velocity
$\Delta J/g$	= ejected gas enthalpy increase (joules/g)
δ	= boundary-layer thickness
ρ_∞	= freestream density

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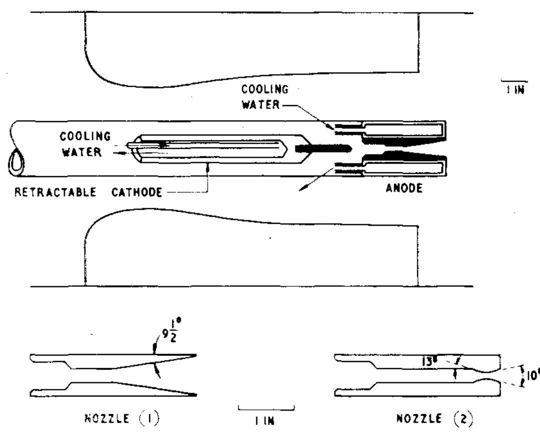

Fig. 1 Arc heater and nozzles.

1. Introduction

The base drag of axisymmetric afterbodies may be significantly reduced by ejecting a small quantity of gas with a low velocity into the "dead-air" region immediately behind the base. Page and Korst[1] have shown both theoretically and experimentally that, for a two-dimensional system, a further but relatively smaller decrease in base drag may be obtained by heating the ejected gas. However, little information appears to be available about axisymmetric afterbodies when the temperature of the ejected gas is raised to that of a conventional rocket propellant. Accordingly, as part of a general program to study the base flow of axisymmetric bodies with gas ejection for possible application to external ballistics, measurements of base pressure have been made on a cylindrical afterbody with the ejected gas heated by an arc heater which increased its enthalpy by up to 3000 joules/g.

2. Apparatus

The measurements were made in a continuously operating supersonic tunnel with an axisymmetric open jet nozzle concentric with a cylindrical centerbody, as described in Ref. 2. For the present tests the centerbody contained an arc heater similar to one used with a low-density wind tunnel[3] (see Fig. 1). The arc heater consists essentially of a central tungsten cathode and a water-cooled copper anode, with internal ducts for cooling water; electrical supplies and pressure tubes are led out through upstream supports to avoid interference with the mainstream flow.

The hot gas ejected from the cylindrical base was argon. Although the tunnel was operated continuously, the argon concentration in the mainstream flow did not rise to more than about 1% because, by the normal operating procedure, the tunnel circuit is continuously evacuated and simultaneously replenished with dry air.

The argon was ejected through either of two nozzles, as shown in Fig. 1. Initially, the precise geometry of the nozzle was not considered important because previous measurements indicated that the increase in base pressure for a given mass flow of ejected gas is independent of nozzle geometry when I is sufficiently small to insure that the ejection velocity is too low to cause entrainment.[4,5] This is the region of practical interest for the reduction of base drag. Accordingly the upstream end of nozzle 1 was made to the same geometry as the anode used previously[3] and the downstream end consisted of a cone of $9\frac{1}{2}°$ semi-angle and exit diameter $\frac{3}{4}$ in. This nozzle was used for all tests up to values of I for which a peak in base pressure occurred ($I \sim 0.003$ for the hot gas). Nozzle 2 (Fig. 1) consisted of an anode with the downstream end in the form of a Mach 2 nozzle (for $\gamma = 1.4$) similar to those used previously[4,5] but scaled to an exit diameter of 0.2 in. This was considered to be typical of the nozzles used in rocket-assisted shells.